

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

C02: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology (DL3)

Assessment Tool Weightage: 30%
Software Cisco Packet Tracer, Wireshark

Code of Conduct

I, A. Alkhaif / 17150258 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

A. Alkhaif
Signature of the student

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Problem Identification	20%	5	2	1	1
Debugging	25%	1	1	1	1
Optimization	20%	1	1	1	1
Analysis	20%	1	1	1	1
Code Quality	15%	1	1	1	1

M/6 Annexure A

Debugging & Optimisation Task

1. A network administrator has configured a workstation with the following settings:

IP Address: 192.168.10.65

Subnet Mask: 255.255.255.192 (/26)

Default Gateway: 192.168.10.1

However, the workstation is unable to communicate with other systems configured in the subnet 192.168.10.0/26

Task

Debug the network configuration by:

- Identifying the subnet to which the configured IP address belongs.
- Determining the network address, broadcast address, and valid host range.
- Verifying whether the configured IP address and gateway are valid.
- Identifying the configuration errors.

IPv4 Debugging

Given:

- IP Address: 192.168.10.65
- Subnet Mask: 255.255.255.192 (126)
- Default Gateway: 192.168.10.1

Ans:

- 1) Subnet: 192.168.10.64/26
- 2) Network Address: 192.168.10.64
- 3) Broadcast Address: 192.168.10.127 ~~126~~
- 4) Valid Host Ranges: 192.168.10.65 - 192.168.10.126
- 5) IP Address valid? = yes.
- 6) Gateway valid? No. (192.168.10.1) belongs to Subnet 192.168.10.0/26, not 192.168.10.64/26
- 7) Configuration Error: Incorrect default gateway.
- 8) optimized configuration.

IP Address: 192.168.10.65

Default Gateway: 192.168.10.65? *

Recommended Gateway: 192.168.10.126

a) usable hosts: 62 ($2^6 - 2$)

1) IPv6 Rebuson

2) IPv6 Debugging.

Device A: 2001:db8::8f00:42:8329/64

Expanded: 2001:db8:0000:000:0000:8f00+0422:8329

Device B: 2001:db8:0:0:1:87/64

Expanded: 2001:db8:0000:

Network Prefix.

Device A: 2001:db8:0:0::/64

Device B: 2001:db8:0:0::/64

Same Subnet:

yes both are in 2001:db8:0:0::/64

Issue:

Addressing is correct: communication failure is likely due to another configuration issue.

Optimized Address.

Device A: 2001:db8:00:10/64

Device B: 2001:db8:0:0:20/64

Available Host Addresses.

$2^{64} \approx 18.4$ quintillion addresses.

subnet optimization

subnet	Network	Host range
/26	¹⁹² 192 .168.10	192.168.1.1 - 192.168.162
/27	192.168.168	192.168.165 - 192.168.176
/28	192.168.196	192.168.197 - 192.168.165.16

Unused Address.

- /26 → 62 useable
- /27 → 30 useable.
- 2/28 → 14 useable

Inefficient Allocation

- Large department should receive /26
- Medium departments should receive /27
- Small department should receive /28

Optimized VLSM

- Largest department → /26
- medium department → /27
- Small department → /28
- Allow remaining space to future department

4) IPv6 Debugging.

Expanded Address 1.

2001:0db8:0000:0000:0000:4f00:0042:8327

Expanded Address B.

2001:0db8:0000:0000:0001:0000:0000:0001

Prefix: 2001:db8:0000::/64

Same Subnet: yes.

Optimized address.

A → 2001:db8:0:0::10/64

B → 2001:db8:0:0::20/64

Available hosts: 2^{64}

5) Routing Table Debugging.

Routing Table.

Destination	Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.1

Destination: 192.168.2.100

Ans: Matching Route: 192.168.2.0/24

Longest prefix match: 192.168.2.0/24

Possible Error: Incorrect or uncatable next hop.

Optimization: verify the route to 10.0.0.3,
remove redundant route and main. operation.
Routing entries.

If no matching Route Exists: Default Route - 0.0.0.0/0